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A p -adic cohomological approach to congruences of meromorphic modular forms

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Numerical observation

Let $C/\mathbb{Q}$: $y^2 + xy = x^3 - x^2 - 2x - 1$, $j(C) = -3375$. Consider the meromorphic modular form

$$f(q) := \frac{E_4(q)}{j(q) + 3375}$$

where $E_4(q) = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n$ and $j(q)$ is the j -invariant.
For every ordinary prime $p > 3$ and every $\ell \geq 0$:

$$a_{np^{\ell+1}}(f) \equiv \mu^2 \cdot a_{np^{\ell}}(f) \pmod{p^{3\ell}}$$

Analytic: $f(q) = \sum_{n \geq 0} a_n(f) q^n$, q -expansion

Arithmetic: μ unit root of $X^2 - a_p(C)X + p$
 $a_p(C) = p + 1 - \#C(\mathbb{F}_p)$

Goal: Describe the geometric meaning of this congruence.

Let $X/\mathbb{Z}_p$ be a *fine smooth moduli scheme* of generalized elliptic curves with extra level structure.

Let $\pi: \mathcal{E} \rightarrow X$ be the universal elliptic curve.

- $\omega := \pi_* \Omega_{\mathcal{E}/X}^1$ Hodge line bundle
- $\mathcal{V} := R^1 \pi_*(\Omega_{\mathcal{E}/X}^\bullet)$ relative de Rham

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$$0 \longrightarrow \omega \longrightarrow \mathcal{V} \longrightarrow \omega^{-1} \longrightarrow 0.$$

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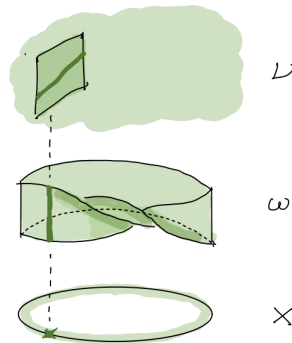
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Fiberwise, for $\alpha \in X(\mathbb{Z}_p)$ non-cuspidal rational point:

$$\begin{aligned} \alpha^* \omega &= \Gamma(E_\alpha, \Omega_{E_\alpha}^1) && \text{global differentials} \\ \alpha^* \mathcal{V} &= H_{\text{dR}}^1(E_\alpha) && \text{de Rham cohomology} \end{aligned}$$

We omitted for simplicity the log-singularities at the cusp



$$M_k(X) = \Gamma(X, \omega^{\otimes k}) \quad \text{modular forms}$$

The relative de Rham cohomology carries a connection:

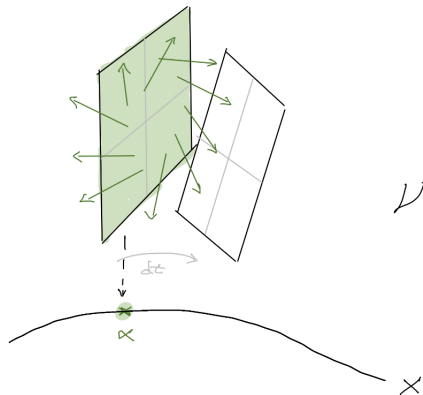
$$\nabla: \mathcal{V} \longrightarrow \mathcal{V} \otimes \Omega_X^1(\log C) \quad \text{Gauss-Manin connection}$$

Extend to higher weight:

$$\mathcal{V}_k := \text{Sym}^k \mathcal{V} \supseteq \omega^{\otimes k}$$

We want to study:

$$H_{\text{dR}}^1(X, \mathcal{V}_k) := \mathbb{H}^1\left(X, \left[\mathcal{V}_k \xrightarrow{\nabla} \mathcal{V}_k \otimes \Omega_X^1\right]\right)$$



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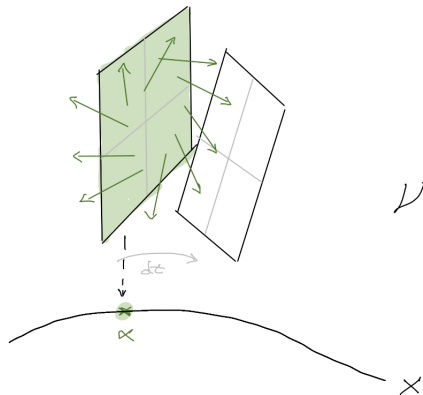
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Eichler–Shimura:

$$0 \rightarrow M_{k+2}(X) \rightarrow H_{\text{dR}}^1(X, \mathcal{V}_k) \rightarrow S_{k+2}(X)^\vee \rightarrow 0$$



Zhang's congruences relate:

- ▶ coefficients of meromorphic forms with **prescribed poles**
- ▶ characteristic polynomial of Frobenius on **cohomology of elliptic curves**

↪ **Residue sequence** for $\alpha \in X(\mathbb{Q}_p)$ corresponding to $E_\alpha/\mathbb{Q}_p$:

$$\begin{array}{ccccccc}
 0 & \longrightarrow & H_{\mathrm{dR}}^1(X, \mathcal{V}_k) & \longrightarrow & H_{\mathrm{dR}}^1(X \setminus \{\alpha\}, \mathcal{V}_k) & \xrightarrow{\mathrm{Res}} & \mathrm{Sym}^k H_{\mathrm{dR}}^1(E_\alpha)(1) \longrightarrow 0 \\
 & & & & \downarrow \wr & & \\
 & & & & M_{k+2}^{\mathrm{mero}, \alpha}(X) / \theta^{k+1} M_{-k}^{\mathrm{mero}, \alpha}(X) & & \theta := q \frac{d}{dq} \text{ Serre operator}
 \end{array}$$

• Middle term $\leftrightarrow$ **meromorphic modular forms**
with pole at α

• Right term $\leftrightarrow$ **cohomology of** E_α , with Frobenius
eigenvalues α_p, β_p

The residue sequence is an exact sequence of **filtered φ -modules**.

$$\begin{array}{ccccccc}
 0 & \longrightarrow & H_{\mathrm{dR}}^1(X, \mathcal{V}_k) & \longrightarrow & H_{\mathrm{dR}}^1(X \setminus \{\alpha\}, \mathcal{V}_k) & \xrightarrow{\mathrm{Res}} & \mathrm{Sym}^k H_{\mathrm{dR}}^1(E_\alpha)(1) \longrightarrow 0 \\
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 \end{array}$$

Goal. Explicitly characterize the **Frobenius action** on $H_{\mathrm{dR}}^1(X \setminus \{\alpha\}, \mathcal{V}_k)$ in terms of **Hecke operators**.

$$U_p f(q) = \sum_{n \geq 0} a_{pn}(f) q^n, \quad V_p f(q) = p^k \sum_{n \geq 0} a_n(f) q^{pn}$$

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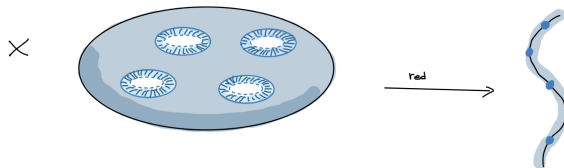
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Problem: The U_p -operator lives on modular curves with **p -level** and **moves around the poles!**

Idea : move to **rigid geometry**. Enlarge the set of poles:

$$D = \{\alpha\} \cup \bigcup_{\bar{\beta} \in \text{SS}} \{\beta\} \quad V_{D,\lambda} = X^{\text{rig}} \setminus \bigcup_{x \in D} B(x, 1 - \lambda)$$



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$$D = \{\alpha\} \cup \bigcup_{\bar{\beta} \in SS} \{\beta\} \quad V_{D,\lambda} = X^{\text{rig}} \setminus \bigcup_{x \in D} B(x, 1 - \lambda)$$

Baldassarri–Chiarellotto '94

$$\begin{array}{ccc} H_{\text{dR}}^1(X \setminus D, \mathcal{V}_k) & \xrightarrow{\sim} & H_{\text{dR}}^1(V_{D,\lambda}, \mathcal{V}_k^{\text{rig}}) \\ \wr \uparrow & & \wr \uparrow \\ M_{k+2}^D(X)/\theta^{k+1} M_{-k}^D(X) & \xrightarrow{\sim} & M_{k+2}^{D,\lambda}/\theta^{k+1} M_{-k}^{D,\lambda} \end{array}$$

$$M_{k+2}^{D,\lambda} := \Gamma(V_{D,\lambda}, \omega^{\otimes k, \text{rig}}) \quad \text{overconvergent modular forms}$$

The **canonical subgroup** allows to raise the level:

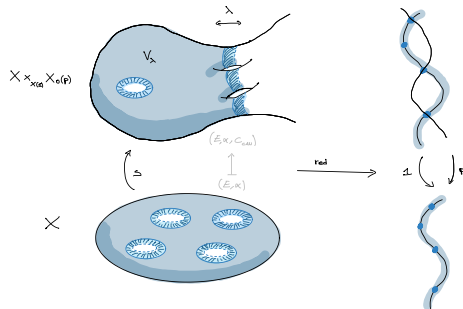
$$E \mapsto (E, C_{E,\text{can}}) \text{ on } V_{D,\lambda}$$

V_p comes from the **moduli lift of Frobenius** $E \mapsto E^{(\varphi)}$ on $V_{D,\lambda}$.

Coleman '96

$$[V_p] = \text{Frob}, \quad [U_p] = \text{Ver}, \quad \text{on } H_{\text{dR}}^1(V_{D,\lambda}, \mathcal{V}_k^{\text{rig}})$$

$$\text{Frob} \circ \text{Ver} = \text{Ver} \circ \text{Frob} = p^{k+1}.$$



$\rightsquigarrow$ the residue sequence is **Frobenius-equivariant**:

$$\begin{array}{ccccccc}
 0 & \longrightarrow & H_{\mathrm{dR}}^1(X, \mathcal{V}_k) & \longrightarrow & M_{k+2}^{D,\lambda} / \theta^{k+1} M_{-k}^{D,\lambda} & \longrightarrow & \bigoplus_{x \in D} \mathrm{Sym}^k H_{\mathrm{dR}}^1(E_x)(1) \longrightarrow 0 \\
 & & & & \curvearrowright U_p, V_p & & \curvearrowright \mathrm{Frob}_p
 \end{array}$$

- ▶ The Frobenius structure on $H_{\mathrm{dR}}^1(X, \mathcal{V}_k)$ is governed by the **Eichler-Shimura isomorphism**
- ▶ The characteristic polynomial of Frobenius on $\mathrm{Sym}^k H_{\mathrm{dR}}^1(E_x)(1)$ has roots $p\alpha^i\beta^{k-i}$ for α, β roots of $X^2 - a_p(E_x)X + p$.

Pulling back a class $[\omega] \in H_{\mathrm{dR}}^1(V_{D,\lambda}, \mathcal{V}_k^{\mathrm{rig}})$ to the cusp $[\infty]$ we obtain an element in $\mathbb{Z}_p[[q]]/\theta^{k+1}(\mathbb{Z}_p[[q]])$.

Theorem (B. 2026)

Fix $E/\mathbb{Q}$ an elliptic curve. Suppose the characteristic polynomial of Frob_p on $\text{Sym}^k H_{\text{dR}}^1(E)$ is coprime with the characteristic polynomial of Frob_p on $H_{\text{dR}}^1(X, \mathcal{V}_k)$.

Then there exist **meromorphic modular forms** $f_0, \dots, f_k \in M_{k+2}^{j(E)}(SL_2\mathbb{Z})$ with poles at $j(E)$ such that

$$a_{mp^\ell}(f_i) \equiv \alpha^i \beta^{k-i} \cdot a_{mp^{\ell-1}}(f_i) \pmod{p^{(k+1)\ell}}$$

where α, β are the roots of the Frobenius polynomial on $H_{\text{dR}}^1(E)$.

Zhang's observation ($k = 2$)

The meromorphic modular form f of weight 4 associated with elliptic curve $C/\mathbb{Q}$ coincides with f_0 satisfying congruence $a_{mp^\ell}(f) \equiv \mu^2 \cdot a_{mp^{\ell-1}}(f_i) \pmod{p^{3\ell}}$.

- ▶ $X = \Gamma \backslash \mathcal{H}_p$ Shimura curve admitting **p -adic uniformization**,
- ▶ Study exact sequences of filtered (φ, N) -modules arising from extension by CM points,
- ▶ Compute Nekovář **local p -adic height pairing** of such extensions,

$$h : \mathrm{Ext}_{\mathrm{MF}^{\mathrm{ad}}(\varphi, N)}(\mathbb{Q}_p, H_{\mathrm{dR}}^1(X, \mathcal{V}_n)) \times \mathrm{Ext}_{\mathrm{MF}^{\mathrm{ad}}(\varphi, N)}(H_{\mathrm{dR}}^1(X, \mathcal{V}_n), \mathbb{Q}_p(1)) \rightarrow \mathbb{Q}_p$$

- ▶ Provide p -adic analytic formula for **higher p -adic Green's functions** $\mathcal{G}_k$,

$$\mathcal{G}_k(z_0, w_0) = \int_{z_0}^{\bar{z}_0} \Psi_k(z, w_0, \bar{w}_0) dz$$

- ▶ We expect the value of $\mathcal{G}_k$ at CM point to be the **p -adic logarithm of an algebraic number**.